Selective Modernization and Bounded Responsiveness in Japan's Waste-Incineration Fleet: A Facility-Level Panel Study

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Abstract

Japan relies heavily on municipal waste incineration, yet a large share of its fleet still burns waste without generating electricity. This creates a two-part modernization problem. Non-generators must first enter power generation, while existing generators must improve conditional performance. Fleet-average studies blur those margins, and generator-only studies miss the entry margin altogether. Using Ministry of the Environment data for FY2005–FY2024, this paper estimates both margins in one national facility panel. It models observed transition into generation among coded facilities first seen without it, then models energy recovery efficiency within a canonical regression frame of operating generators. Transition is selective rather than diffuse: facilities older than 0–10 years are less likely to record transition in the next observed year, while larger facilities are more likely to do so. Among operating generators, efficiency is lower at older plants and higher at larger, more fully utilized ones, while between-facility heterogeneity dominates within-facility movement. An aggregate view therefore understates both the selectivity of entry and the persistence of performance hierarchy. These patterns are descriptive within the linked samples, not causal estimates of a single modernization mechanism. For municipal fleet planning, non-generators and mature generators should not be managed as one average segment.

Keywords: waste incineration; waste-to-energy; Japan; energy recovery; facility panel; transition

1 Introduction

Japan operates one of the world's most incineration-dependent municipal waste systems, yet many facilities still burn waste without generating electricity from the heat they produce (Ministry of the Environment Japan, 2022; Uno, 2015; Tabata & Tsai, 2016; Sakai et al., 2011). This is not a marginal technical detail. For decades, Japanese municipal waste governance has relied on thermal treatment for hygienic disposal, volume reduction, and local waste autonomy, while limited landfill space and strict environmental controls pushed municipalities toward sophisticated incineration infrastructure (Brunner & Rechberger, 2015; Sakai et al., 2011). Sectoral planning now treats the remaining electricity-generation gap as part of the waste sector's decarbonization challenge (Yamada et al., 2023; Greenhouse Gas Inventory Office of Japan & Ministry of the Environment Japan, 2024). The transition problem is therefore not whether Japan uses incineration. It is which facilities move into useful energy recovery and how well they perform after they enter.

Aggregate fleet summaries flatten two distinct questions. One is extensive: which facilities record observed transition into power generation at all? The other is intensive: among plants that already generate, which facilities achieve high energy recovery efficiency? The system can therefore look uniformly slow in the mean while combining selective entry at one end of the fleet with persistent performance hierarchy at the other.

This paper estimates both margins in one national facility-level panel. Its contribution is empirical and design-based: the same linked data structure is used to estimate observed transition into generation and conditional performance within generation. Within the coded at-risk frame, observed transition is selective toward younger and larger facilities. Within the canonical regression frame, efficiency is strongly structured by age, scale, and utilization, while within-facility movement remains limited relative to between-facility heterogeneity. The same fleet can therefore look merely slow in aggregate while containing two different problems: selective entry into generation and bounded performance among generators. For municipal planning, that distinction changes the first diagnostic step. Non-generators and mature generators should be evaluated as different asset-management questions before they are summarized as one fleet.

The gap addressed here is narrower than a generic claim that Japan has been understudied. Waste-to-energy research can describe fleet trajectories, and generator-only studies can explain conditional performance once plants already operate as generators. What remains uncommon is one linked municipal-fleet analysis that estimates both margins and asks whether they point to the same modernization bottleneck. Japan makes that contrast visible because persistent non-generators, old small plants, and a more modern generating segment coexist in the same administrative system. In such mixed fleets, adoption and conditional performance should be modeled separately before they are interpreted together.

The rest of the paper proceeds as follows. Section 2 positions the paper against the literature most relevant to the analytical split. Section 3 introduces the two linked analytical frames and the main estimation choices. Section 4 reports the adoption and efficiency results in sequence, then ties them together. Section 5 interprets the combined finding, explains what the data still cannot identify, and states a short set of evidence-consistent implications. Section 6 concludes.

2 Literature Positioning

This paper speaks to three literatures that often meet in practice but start from different empirical positions: waste-to-energy systems, facility-level efficiency analysis, and infrastructure lock-in. The review is organized around the paper's central split between entry into generation and conditional performance after entry. The relevant gap is not simply that Japan needs another case study. It is that existing studies often describe fleets in aggregate or explain generator performance conditional on operation, yet rarely estimate both margins in one linked municipal-fleet design.

Work on waste-to-energy systems often documents national trajectories, technology choices, or lifecycle implications of thermal treatment (Astrup et al., 2009; Astrup et al., 2015; Brunner & Rechberger, 2015; Sun et al., 2018). Japan-specific work has mostly emphasized technology upgrading, heat-use constraints, and sectoral decarbonization scenarios rather than facility-level transition modeling (Uno, 2015; Tabata & Tsai, 2016; Yamada et al., 2023). Policy-facing work reaches a similar conclusion from a different angle: waste-to-energy is treated as useful only when embedded in a wider waste hierarchy and resource-recovery strategy rather than as a stand-alone justification for thermal treatment (European Commission, 2017; Sakai et al., 2011). This literature is indispensable for understanding why energy recovery matters, but it usually treats the incineration fleet as a sectoral category. It can show whether energy recovery is expanding, whether heat supply remains constrained, or whether the sector matters for net-zero planning. It is less well suited to distinguishing facilities that enter generation from those that do not.

Facility-level efficiency studies are closer to the present design, but they typically begin after entry has already occurred. Studies in Taiwan, for example, evaluate operating incinerators by decomposing waste-treatment, electricity-generation, or revenue performance within existing plants (Chen et al., 2012; Yeh, 2020). Other work focuses on energy-recovery criteria, plant scale, heat use, and the system consequences of different waste-to-energy configurations (Grosso et al., 2010; Münster & Meibom, 2010). Recent Chinese plant-level work is also highly informative about performance differentials inside the generating segment and the effectiveness of upgrading strategies at scale (Cui et al., 2026; Liu et al., 2025; Han et al., 2025). Those studies help define the intensive-margin question. They do not directly answer which non-generating facilities enter generation in the first place.

The lock-in literature adds a different expectation. Infrastructure performance may be shaped by durable design choices, inherited scale, and institutional arrangements rather than by frequent late-life reversals at mature facilities (Unruh, 2000; Geels, 2004; Seto et al., 2016). The useful implication here is not that incineration plants are literally irreversible. It is that cross-facility differences may matter more than repeated performance resets. In municipal infrastructure, that persistence can also be institutional rather than purely technical: plant networks are shaped by jurisdictional boundaries, merger histories, charging regimes, and the political difficulty of reorganizing service territories (Rausch, 2006; Sakai et al., 2008; Sakai et al., 2011). That expectation is only partially visible if entry into generation and performance within generation are never separated.

Read together, the literatures imply a practical identification problem for fleet studies. Fleet-level work can show aggregate progress but cannot tell whether low average performance reflects many non-generators, weak generator performance, or both. Generator-only work can estimate efficiency correlates after entry, but not who stays outside generation. Lock-in work explains why mature infrastructure may remain stratified, but not whether the key margin lies before or after entry. The contribution of this paper is to use one linked panel to expose those blind spots without treating them as one average process.

3 Data and Design

The analysis uses the Ministry of the Environment's General Waste Treatment Survey for FY2005–FY2024 (Ministry of the Environment Japan, 2022). From 23,599 facility-year rows, the coded frame retains 19,827 observations across 2,948 identifiable facilities. The paper then separates two linked samples because one sample cannot answer both parts of the transition problem. Those frames are not only data filters. They are the analytic structure that keeps observed entry into generation separate from conditional performance after entry. The survey is useful for this purpose because it covers both generating and non-generating facilities inside the same administrative system. That makes it possible to ask a question that many sector studies cannot ask cleanly: which facilities record observed transition into generation, and how do facilities perform once they are already inside the generating segment?

The first frame is the coded adoption frame. It includes facilities first observed without power generation and follows them until they either record observed transition into generation or remain non-generating in the panel window. After excluding left-censored facilities already generating in their first observed year, the adoption risk set contains 13,770 facility-years across 2,035 facilities, with 141 observed first-adoption events.

The main adoption model is a lagged discrete-time logit hazard estimated on 11,717 observations across 1,915 facilities and 140 retained events. Predictors are prior-year age band and prior-year design capacity, with year fixed effects, prefecture fixed effects, and facility-clustered standard errors. This is an observed-transition model, not a complete structural model of all possible modernization pathways. The design follows grouped event-history logic: each coded facility-year contributes to the risk set until first event occurrence (Allison, 1982; Beck et al., 1998). That distinction matters because the paper is not estimating a continuous engineering retrofit process. It estimates the probability that a facility first records entry into power generation in the next observed year, conditional on still being at risk. The lagged predictor structure ensures that age band and capacity are measured before the observed event rather than on the event row itself.

The second frame is the canonical generator frame. It contains operating facilities with positive throughput and positive power output, after standard cleaning and efficiency bounding. The operating-generation sample contains 6,660 rows before identifier and regression cleaning. Of those, 907 rows lack official facility codes and are excluded from the canonical regression frame, leaving 5,683 observations across 1,016 facilities. The dependent variable is winsorized log electricity generated per tonne processed. Main predictors are facility age, design capacity, capacity utilization, waste heating value, and a grid-emission control. The main specifications are pooled OLS, year-FE, RE, and year-FE plus RE models, all used as structured descriptive models rather than clean structural estimates (Wooldridge, 2010).

The efficiency results should therefore be read as conditional patterns within the canonical regression frame, not as estimates for the entire generating fleet. This frame is intentionally narrower than the operating-generator universe because the argument depends on facility-level comparison over time. Rows without official facility codes cannot support that comparison. The frame is better understood as the canonical identifiable generator sample than as a census of all generation activity. The paper also uses a bounded efficiency metric because the empirical question is not boiler thermodynamics in isolation, but administrative performance in electricity recovered per tonne processed. That puts the paper closer to applied energy-recovery studies than to plant-level engineering optimization alone (Grosso et al., 2010; Münster & Meibom, 2010).

Two administrative-data checks are reported in the supplement. First, a small set of official facility codes appears more than once within the same fiscal year, so a composite-ID sensitivity appends facility names to affected duplicate codes. Second, heating value is treated as a noisy control and is checked under plausible-value restrictions. Neither sensitivity changes the main sign pattern or the substantive interpretation.

The two layers belong in one paper because they answer sequential parts of the same modernization problem. The adoption layer identifies which facilities appear to enter the generating regime. The efficiency layer identifies whether large performance gaps remain once facilities are already inside that regime. Without the first layer, the paper would reduce transition to generator performance alone. Without the second, it would say who enters generation but not whether major efficiency differences remain inside the generating segment. The two samples are linked but non-identical, so they should not be read as one causal pathway. They are instead the extensive-margin gate and

Table 1: Linked analytical framework

Margin	Linked sample	Empirical question	Paper role
Adoption margin	Coded at-risk frame: 13,770 facility-years, 2,035 facilities, 141 observed first-adoption events	Which facilities record observed transition into generation?	Shows whether entry into generation is selective rather than diffuse
Efficiency margin	Canonical generator frame: 5,683 observations across 1,016 operating generators	How does performance vary once generation already exists?	Shows whether mature generator performance remains bounded
Synthesis	Two linked but non-identical analytical frames	Would one average-fleet view misstate the modernization bottleneck?	Shows why entry and mature performance should not be read as one average process

Note: the adoption margin is estimated with a lagged discrete-time hazard. The efficiency margin is estimated with descriptive pooled, year-FE, and RE panel specifications.

the conditional-performance layer of the same modernization problem.

This linked design also clarifies what the paper is not trying to do. It is not estimating a single structural law of fleet modernization, and it is not asking whether one estimator dominates all others in abstract econometric terms. Instead, it asks what can be learned when a municipal fleet is partitioned into the margin where generation first appears and the margin where generating plants continue to differ. That framing matters for interpretation. If the extensive margin looks selective while the intensive margin remains hierarchical, then the relevant practical conclusion is not that the fleet is uniformly lagging. It is that different parts of the fleet face different modernization tasks.

The main identification limits are explicit. The design is a diagnostic fleet decomposition, not a policy-effect estimator. In the adoption layer, the paper models observed transition within the coded risk set, not unrestricted fleet-wide modernization. In the efficiency layer, age is closely tied to time and within-facility movement is limited, so the defended interpretation is one of structured conditional association rather than strict causal identification. All reported effects should be read as conditional associations within the specified samples, not as estimates of structural causality or policy effects.

The paper also does not claim that the low within-facility variance ratio resolves all fixed-effects concerns. It uses that variance structure to explain why cross-facility descriptive comparison remains substantively useful for the question at hand. This is where the paper differs from a methods-first estimator comparison. The question is not whether fixed effects can be forced in, but whether cross-facility description remains informative under explicit sample limits. The answer defended here is yes: the variance structure and stable sign pattern make the descriptive models useful, but not structural.

4 Results

4.1 Adoption into generation is selective rather than diffuse

The adoption results show a strongly selective transition pattern. In the risk set, annual event rates collapse after age 10 and rise sharply across capacity quartiles. Facilities aged 0–10 years account for 102 first-adoption events, while the three older age bands together account for only 39. By capacity, the largest quartile accounts for 99 first-adoption events, whereas the smallest quartile records only 1.

The discrete-time logit hazard summarizes the same pattern in average marginal effects. Relative to 0–10 year facilities, plants aged 10–20 years are about 1.76 percentage points less likely to record transition in the next observed year, plants aged 20–30 years are about 1.72 percentage points less likely, and plants aged 30 years or more are about 1.13 percentage points less likely. Each additional 100 t/day of prior-year design capacity raises annual transition probability by about 0.50 percentage points. The sign pattern is stable in the complementary log-log and linear probability robustness variants.

These effects should be read within the coded at-risk frame, not as a model of all modernization activity in the Japanese fleet. Even within that narrower frame, the evidence does not show broad late-life conversion among old, small plants. Transition is concentrated among facilities that were already younger and larger before the event year. Older plants do transition in some cases, but they do so less often and do not define the main event pattern. The extensive margin therefore looks like selective modernization rather than broad catch-up across the whole fleet. This matters because a descriptive fleet mean could be read as gradual modernization delayed by inertia, when the event pattern is more selective than gradual.

The pathway audit supports that interpretation without overstating mechanism. Among the 141 observed adoption events, 82 are classified as reset- or rebuild-like, 38 as continuity-type upgrades, 20 as forward-dated or placeholder entries, and 1 as unresolved. This is descriptive pathway evidence, not mechanism identification. The event mix is more consistent with capital-intensive pathways than with diffuse late-life catch-up, but it does not uniquely identify replacement, major refurbishment, or new build as the single pathway. In the main text, the audit therefore functions as a credibility guard rather than as a coequal source of originality.

That distinction keeps the adoption result from becoming a story of technological inevitability. The event pattern does not imply that older plants never upgrade, nor that small facilities have no role in local waste management. It implies something narrower: within the coded at-risk frame, recorded entry into generation is not centered in the segment most likely to be described as lagging in simple fleet summaries. The modernization margin visible in the data is selective from the start.

4.2 Performance within generation is bounded and strongly structured

Within the canonical regression frame, the strongest descriptive result is the variance structure. The within-to-total variance ratio of pooled log-efficiency is 0.1499, meaning that most variation is between facilities rather than within facilities over time. The ratio remains low in both the pre-Fukushima and post-Fukushima windows, falling from 0.1795 before 2011 to 0.0956 after 2011. This does not prove irreversibility. It does show that mature generators do not frequently undergo

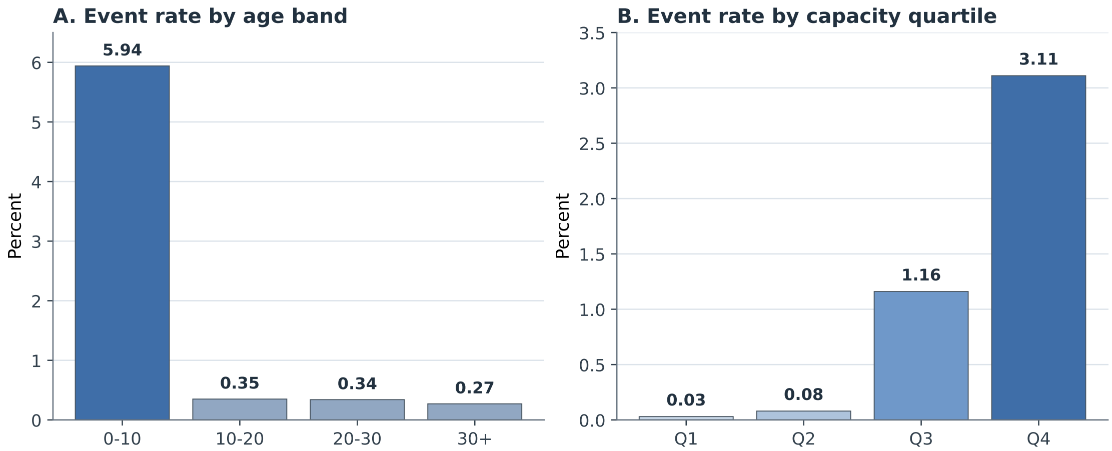

Figure 1: Observed adoption event rates by age band and capacity quartile in the coded at-risk frame.

Table 2: Main lagged hazard results for observed transition into generation

Variable	AME (pp)	SE (pp)
Prior-year age 10–20 yrs (vs 0–10)	-1.76	0.28
Prior-year age 20–30 yrs (vs 0–10)	-1.72	0.42
Prior-year age 30+ yrs (vs 0–10)	-1.13	0.39
Prior-year capacity (per 100 t/day)	0.50	0.20
Model summary		
Observations		11,717
Facilities		1,915
First-adoption events		140
Pseudo-R-squared		0.1842

Note: entries are average marginal effects in percentage points from the main lagged logit hazard with year and prefecture fixed effects and facility-clustered standard errors.

large late-life movements that reshape the fleet distribution. It also does not isolate vintage effects from all other durable plant characteristics. More narrowly, it supports cross-facility descriptive comparison rather than clean causal isolation of vintage itself.

The coefficient patterns point in the same direction. Efficiency is consistently lower at older facilities and higher at larger and more fully utilized ones. The magnitudes differ across models, but the sign pattern is stable. The emphasis is therefore on structured conditional association rather than on any single structural parameter. This is consistent with earlier facility-level work showing that energy recovery performance is uneven across operating incinerators and that plant scale and operational intensity matter for output performance (Chen et al., 2012; Yeh, 2020; Grosso et al., 2010). What this paper adds is the linked comparison to the non-generating segment: the same fleet that shows selective entry at one margin also shows a stable hierarchy among mature generators at the other.

The efficiency margin therefore looks bounded rather than static. Facilities can respond through utilization and operational discipline, but age and scale hierarchies remain strong. This is where the paper diverges from a simple engineering-upgrade narrative. Recent large-scale Chinese studies show that substantial gains can still be unlocked through technology upgrades, pollutant control,

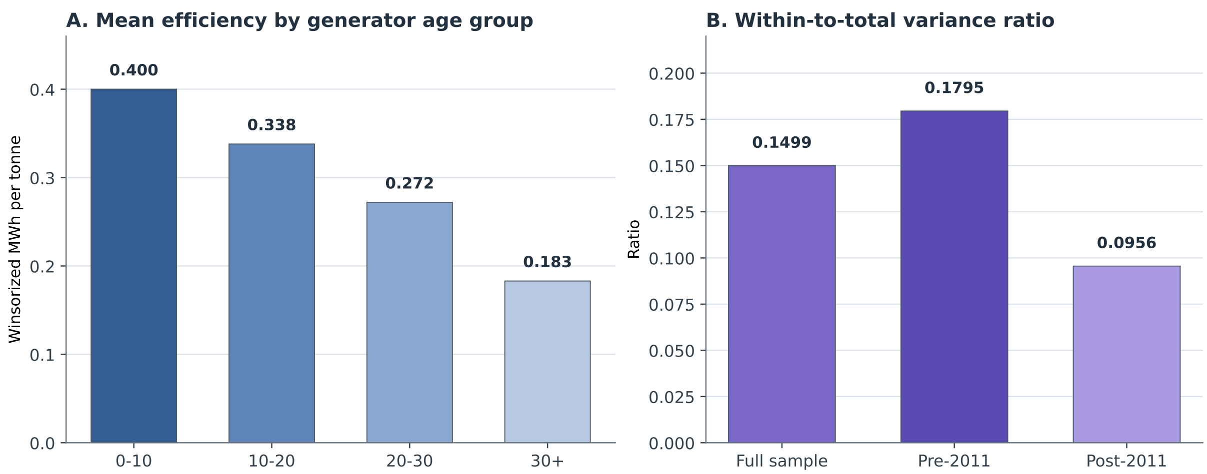

Figure 2: Mean efficiency declines across generator age groups, while the within-to-total variance ratio stays low in the full sample and in pre/post-Fukushima splits.

Table 3: Core conditional-efficiency specifications in the canonical generator frame

Variable	Model 1 Pooled OLS	Model 2 Year FE	Model 3 RE	Model 4 Year FE + RE
Facility age (years)	-0.0279***	-0.0348***	-0.0188***	-0.0332***
	(0.0022)	(0.0022)	(0.0025)	(0.0021)
Capacity (100 t/day)	0.0874***	0.1030***	0.0405***	0.0519***
	(0.0083)	(0.0086)	(0.0083)	(0.0096)
Capacity utilization	0.7468***	0.7789***	0.6199***	0.5411***
	(0.1421)	(0.1346)	(0.0997)	(0.0943)
Heating value (MJ/kg)	0.0010	0.0032	0.0006	0.0012
	(0.0023)	(0.0021)	(0.0012)	(0.0010)
Grid EF (kg-CO_2/kWh)	0.3182	-0.4466	1.6333***	-0.1951
	(0.2219)	(0.2714)	(0.1965)	(0.2101)
Observations	5,683	5,683	5,683	5,683
Facilities	1,016	1,016	1,016	1,016
R-squared	0.2470	0.3721	0.1647	0.3076

Note: standard errors are in parentheses. *** $p < 0.01$. Coefficients are reported as structured conditional associations rather than as strict structural parameters.

waste classification, and load-rate improvements, but they do so within already-generating systems rather than at the point of first entry (Liu et al., 2025; Han et al., 2025). The present results are consistent with bounded improvement rather than natural convergence after entry.

This is why the intensive margin cannot be inferred from the adoption margin alone. A facility can be inside generation without being close to the frontier. At the same time, operations alone are unlikely to erase large vintage and scale gaps once plants are mature. Conditional performance is therefore a bounded-performance problem rather than a simple continuation of the entry problem.

For interpretation, this matters as much as the adoption result. If the paper only documented selective entry, a reader could infer that the main task was simply to push more facilities into generation. If it only documented generator hierarchy, a reader could infer that non-generators were just lagging versions of the same problem. The linked result rejects both shortcuts. The fleet appears divided between a segment that still struggles to enter generation and a segment where entry has occurred but performance remains uneven and bounded.

4.3 Why the two results belong together

Read together, the two margins change the modernization story. The adoption results show that entry into generation is already selective before conditional efficiency is considered, while the efficiency results show that large performance gaps inside the generating segment are not easily erased through within-facility movement alone. A one-average-fleet model would flatten those margins into a single modernization narrative and would therefore understate both the selectivity of entry and the persistence of cross-facility performance differences. The point is not that the two samples form one strict causal chain, but that they identify different constraints within the same fleet. One concerns who gets into generation at all; the other concerns how far operating generators can move once they are already there.

5 Discussion

The paper's main interpretive claim is methodological: in this fleet, entry into generation and performance within generation are linked but distinct estimands. Modeling them separately shows that the weakest part of the fleet and the mature generating segment are constrained in different ways. This matters because municipal fleets often contain both non-generators and mature generators at the same time. If those segments are averaged together, the analyst sees only a muted fleet mean rather than the combination of selective entry and persistent hierarchy that structures the system.

The substantive interpretation is correspondingly two-part. On the adoption margin, the data do not support broad late conversion among old small plants. Observed transition within the coded at-risk frame is concentrated among younger and larger facilities, and the pathway audit is more consistent with a capital-intensive event mix than with diffuse late-life catch-up. On the efficiency margin, age, scale, and utilization still matter strongly within the canonical regression frame, while within-facility movement remains modest relative to the cross-sectional hierarchy. Read together, the evidence points more toward selective entry into the generating regime and bounded responsiveness within generation than toward easy convergence once entry has occurred.

The interpretation has clear limits. The pathway audit does not prove that replacement is the unique pathway of modernization, and the regressions do not provide strict causal estimates of vintage lock-in or clean estimates for all operating generators in Japan. Alternative interpretations remain possible, including reporting compression, unobserved retrofit histories, unmeasured governance differences, and institutional constraints that limit operational responses. The defended claim is therefore narrower: the data support a selective modernization process and a bounded performance envelope, not a uniquely identified mechanism or a full causal hierarchy.

For the weakest segment, especially older non-generators and small plants, the evidence points more toward capital-renewal planning than diffuse late-life operational improvement. For the already-generating segment, utilization, routing, and selective upgrading remain real levers, but they appear more likely to preserve or modestly improve performance within the existing envelope than to eliminate large inherited gaps. For municipal waste planners, the practical takeaway is to separate fleet triage from generator optimization: first identify which non-generators plausibly warrant renewal, then identify which operating generators still have room for incremental gains.

That distinction is administrative as well as technical. Japanese waste systems are organized through municipalities whose planning boundaries do not always align with efficient waste sheds, and intermunicipal reorganization has its own political costs and institutional legacies (Rausch, 2006; Sakai et al., 2008). In that setting, collapsing non-generators and generators into one average segment risks hiding the difference between asset renewal, which is lumpy and governance-heavy, and generator optimization, which is more incremental and operational.

The broader policy context points in the same direction. Comparative waste policy studies and European framework discussions both treat waste-to-energy as valuable only when embedded inside a wider hierarchy that preserves prevention, reuse, and recycling priorities (Sakai et al., 2011; European Commission, 2017). The empirical split helps explain why those hierarchies matter. A municipal system can have real energy-recovery gains available at the non-generating margin while still facing a different, narrower set of decisions inside the already-generating segment.

For climate interpretation, the relevant question is not simply whether a plant generates, but what kind of generator it is and how much improvement remains inside that segment. Waste-to-energy performance is judged against both an internal engineering standard and the emissions profile of the broader energy system, including the avoided-emissions logic built into carbon accounting (Astrup et al., 2009; Münster & Meibom, 2010). In Japan's current decarbonization setting, where national inventories and scenario work track sectoral emissions and energy mix changes, the two-part design is useful because it keeps those questions separate (Greenhouse Gas Inventory Office of Japan & Ministry of the Environment Japan, 2024; Yamada et al., 2023).

That separation is also what makes the paper more understandable for a planning audience. Municipal systems rarely choose among abstract technological ideals. They decide whether to renew an aging non-generator, coordinate waste routing toward a larger plant, maintain an existing generator, or invest in an upgrade for a plant that already produces electricity. Those are related decisions, but they are not interchangeable. Keeping the extensive and intensive margins separate helps planners locate the bottleneck without relying on one blended fleet mean.

6 Conclusion

Japan's incineration transition is not one smooth modernization process. Within the coded adoption frame, observed entry into generation is selective rather than diffuse. Within the canonical generator frame, performance remains stratified by age, scale, and utilization, with limited within-facility movement relative to between-facility differences. Read together, those margins show why an aggregate fleet view can misstate the modernization bottleneck. The paper does not identify one unique pathway or intervention hierarchy, but it does show why municipal fleet studies gain by separating adoption from conditional performance. Municipal fleet planning should treat adoption and generator optimization as separate tasks rather than one average problem.

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CRediT Authorship Contribution Statement

Pann Phetra: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Visualization, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethical Approval

This study uses publicly released administrative facility data and does not involve human participants, animal subjects, or private personal data.

Data Availability

The facility-level source data are derived from the Ministry of the Environment Japan General Waste Treatment Survey, which is publicly released by the Ministry of the Environment. The cleaned analysis outputs, figure-generation scripts, manuscript figures, and reproducible paper workspace can be made available by the author on reasonable request, subject to any redistribution limits attached to the source administrative files.

Generative AI and AI-Assisted Technologies Statement

During the preparation of this manuscript, the author used OpenAI Codex and Anthropic Claude to support drafting, language revision, and organizational planning. These tools were not used as authors, did not generate or verify the underlying data, and did not replace author review. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the manuscript.

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